Lab 2 - GradMap Prototype Product Specification

Team Ruby

Old Dominion University Department of Computer Science

CS 411W Professional Workforce Development II, Spring 2026

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Apr 25, 2026

Version 2

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3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces (O: Dani)

The system shall allow users to input and retrieve information through a web-based interface. The system will accept user inputs such as login credentials, course selections, and feedback messages, and process this data to generate outputs including dashboards, academic schedules, validation alerts, and advisor feedback. All inputs shall be validated before processing, and outputs shall be displayed clearly and in real time.

3.1.1.1 User Authentication (O: Brice)

The User Authentication module is responsible for verifying the identity of individuals accessing the GradMap application and ensuring secure session management. This module interacts directly with the Users database table to validate credentials and enforce role-based access control for Students, Advisors, and Administrators. All passwords must be securely hashed, and authenticated sessions will be maintained using industry-standard methods.

3.1.1.1.1 Login Page

3.1.1.1.1.1

The system shall provide a user interface containing input fields for username and password, along with a “Login” submission button

3.1.1.1.1.2

The system shall verify the provided credentials against the hashed passwords stored in the Users table.

3.1.1.1.1.3

If authentication is successful, the system shall generate a JWT containing the user’s ID and role, and establish a secure user session.

3.1.1.1.1.4

Following a successful login, the system shall redirect the user to the appropriate landing page based on their role.

3.1.1.1.2 Registration Page

3.1.1.1.2.1

The system shall provide a user interface for account creation requiring the following fields: First Name, Last Name, Username, Password, Confirm Password, and Role Selection.

3.1.1.1.2.2

The system shall verify that the requested username does not already exist within the Users table to prevent duplicate accounts

3.1.1.1.2.3

The system shall create a new record in the User's table containing the username, password_hash, and selected role

3.1.1.1.2.4

Upon successful registration, the system shall display a confirmation message and redirect the user to the Login Page

3.1.1.2 Student Interface (O: Ivan M: Brice)

The Student Interface provides all functionality available to student users after successful authentication (3.1.1.1). This interface allows students to manage their academic planning process through dashboards, schedule planning tools, and course exploration features. The interface is implemented through multiple frontend components, including StudentDashboard.jsx, GradPlansPage.jsx, MyAvailabilityPage.jsx, and CourseCataloguePage.jsx fileciteturn3file0.

3.1.1.2.1 Dashboard - Student (O: Ivan)

The Student Dashboard serves as the main landing page for student users after login.

3.1.1.2.1.1 The system shall display student profile information, including:

- Name
- Major
- GPA
- Academic progress

3.1.1.2.1.2 The system shall display alerts and notifications related to:

- Plan updates
- Advisor feedback
- Time Conflicts

3.1.1.2.1.3 The system shall display a summary of the student's current GradPlans

3.1.1.2.1.4 The system shall provide navigation controls to:

- GradPlans Page (3.1.1.2.2)
- MyAvailability Page (3.1.1.2.3)
- Course Catalogue Page (3.1.1.2.4)
- References Page (3.1.1.2.5)

3.1.1.2.1.5 The system shall organize dashboard content into clearly defined sections for readability

3.1.1.2.2 GradPlans Page (O: Ivan)

The GradPlans Page allows students to view, create, and modify their academic schedules.

3.1.1.2.2.1 The system shall display all GradPlans associated with the student with the following information:

- Semester
- Selected courses
- Total credit hours
- Plan status (Draft, Pending, Approved, Needs Revision)

3.1.1.2.2.2 The system shall allow students to:

- Add courses to a plan
- Remove courses from a plan
- Submit plans for advisor review

3.1.1.2.2.3 The system shall update plan data in real time after modifications

3.1.1.2.2.4 The system shall validate plan constraints before saving

3.1.1.2.3 MyAvailability Page (O: Ivan M: Brice)

The MyAvailability Page allows students to define their weekly availability for scheduling purposes.

3.1.1.2.3.1 The system shall display a weekly schedule grid

3.1.1.2.3.2 The system shall allow students to input availability by:

- Selecting days of the week
- Selecting time blocks

3.1.1.2.3.3 The system shall allow students to:

- Add availability blocks
- Remove availability blocks

3.1.1.2.3.4 The system shall store availability data for use in schedule generation (3.2.1.2)

3.1.1.2.3.5 The system shall validate availability inputs before saving

3.1.1.2.4 Course Catalogue Page

The Course Catalogue Page allows students to browse and select courses.

3.1.1.2.4.1 The system shall display a list of available courses with the following information:

- Course name
- Course code
- Credits
- Description
- Prerequisites

3.1.1.2.4.2 The system shall allow students to:

- Search for courses
- Filter courses
- Add courses to a GradPlan

3.1.1.2.4.3 The system shall retrieve course data from the database

3.1.1.2.5 References Page

The References Page provides students with access to additional academic resources.

3.1.1.2.5.1 The system shall display links to external and internal resources

3.1.1.2.5.2 The system shall allow students to navigate to reference materials

3.1.1.2.5.3 The system shall organize resources into categories for usability

3.1.1.3 Advisor Interface (O: Matthew)

The Advisor Interface module is responsible for allowing an advisor to access all information and functions necessary for them to complete their responsibilities. This module interacts directly with the student database table to display a student's course information to an advisor. All information about a student will be restricted to only the advisor in charge of that student.

3.1.1.3.1 Home Page

The home page serves as the landing page for all advisors after logging in. It allows advisors to view all the features that they have access to. It will display the students an advisor is in charge of, the student's graduation status, and a list of any scheduling plans submitted by a student.

3.1.1.3.2 Schedule Review

By clicking on one of the submitted plans in the submitted schedules list, the advisor will be granted access to information about the student and their plan. This will display the plan's current status to the advisor along with a summary of the plan and any alerts about the plan. There will be a section that will display the plan in full to see what a student has decided on with a messaging system attached to the student's plan. There is also an option to approve or reject a student's plan.

3.1.1.3.3 Advisor Feedback

The messaging system appears when a student's plan is selected. This message system is connected directly to the student's plan and will immediately display any feedback an advisor provides about the student's plan for the student to view on the page with their plan.

3.1.1.4 Admin Interface (O: Jackson)

The Admin Interface serves to allow server administrators to modify or edit school-provided data within the system. After passing through the User Authentication Module (See Section 3.1.1.1), Admins will be able to directly modify the internal information database through MySQL..

3.1.1.4.1 Home Page

The Home Page shall allow for navigation and account management for the Administrators. It will display account information and provide links to the other two main pages.

3.1.1.4.2 Assignment Page

The Assignment Page shall allow the administrator to modify the database directly, as well as what accounts can view what information. The database will need to be saved before it is uploaded.

3.1.1.4.3 Read Along Page

The Read Along Page shall allow administrators to view the application through the lens of how an Advisor or Student would see it, ensuring that students with certain qualifications are given the correct information, and that advisors are assigned to the correct students.

3.2 Functional Requirements

3.2.1 Algorithms (O: Dani)

The GradMap system shall use algorithms to process user inputs and generate accurate academic schedules. The system shall analyze course selections, prerequisites, availability, and time constraints to create valid schedules and detect conflicts. It shall also handle account management functions such as login verification and account creation.

3.2.1.1 Account Management (O: Ivan)

The Account Management module handles user registration, authentication, and session control within the GradMap system.

3.2.1.1.1 Account Creation

3.2.1.1.1.1 The system shall accept user input from the registration page (3.1.1.1.2)

3.2.1.1.1.2 The system shall require:

- Username
- Password
- Role

3.2.1.1.1.3 The system shall validate all input fields

3.2.1.1.1.4 The system shall ensure usernames are unique

3.2.1.1.1.5 The system shall hash passwords before storing them

3.2.1.1.1.6 The system shall store user data in the database

3.2.1.1.1.7 The system shall create role-specific profiles

3.2.1.1.1.8 The system shall return success or failure messages

3.2.1.1.2 Login

3.2.1.1.2.1 The system shall accept login credentials from the login page (3.1.1.1.1)

3.2.1.1.2.2 The system shall authenticate credentials against stored data

3.2.1.1.2.3 The system shall deny access if credentials are invalid

3.2.1.1.2.4 The system shall display an error message for failed login attempts

3.2.1.1.2.5 The system shall generate a session token upon successful login

3.2.1.1.2.6 The system shall redirect users to their respective dashboards:

- Student Dashboard (3.1.1.2.1)
- Advisor Dashboard
- Admin Dashboard

3.2.1.2 Schedule Creation/Generation (O: Brice)

The Schedule Creation module is the core functional component of the Student Planning engine within GradMap. It allows student users to build, modify, and visualize their academic trajectory by assigning specific courses to future semesters. This module interacts heavily with the Plans, Planned_Courses, Courses, Time_Slots, and Availability tables to ensure accurate and actionable academic progress.

3.2.1.2.1 Course Selection and Assignment

3.2.1.2.1.1

The system shall provide a search and filtering interface allowing users to browse the Courses table by course_code, course_name, department, and category.

3.2.1.2.1.2

The system shall allow a student to create a new academic plan, generating a unique record in the Plans table associated with their student_id.

3.2.1.2.1.3

The system shall enable users to select a course and assign it to a specific semester and year within their active plan.

3.2.1.2.1.4

The system shall allow users to easily remove or move a planned course to a different semester.

3.2.1.2.2 Schedule Validation and Conflict Resolution

3.2.1.2.2.1

When a user attempts to add a course, the system shall query the Prerequisites tables to verify if the student has already planned or completed the required prerequisite_course_id(s) in prior semesters.

3.2.1.2.2.2

If a prerequisite requirement is not met, the system shall display an error message detailing the missing prerequisite and prevent the course addition.

3.2.1.2.2.3

If a time conflict is detected, the system shall alert the user and prompt them to select an alternative course or section.

3.3 Performance Requirements

3.3.1 Application Performance (O:Dani)

The GradMap system shall provide responsive performance for all users. Login, dashboard, and schedule pages shall load within 2 seconds under normal conditions. Schedule generation and validation shall be completed within 2 seconds. The system shall support at least 100 concurrent users without major delays.

3.4 Logical Database Requirements (O: Brice)

This section outlines the logical data models, data storage requirements, and data management principles for the GradMap system. The database is the central repository for all application data, ensuring that user information, academic catalogs, and student plans are securely stored, linked, and available.

3.4.1 Data Management

Data management encompasses how information is structured, processed, and maintained within GradMap. The system handles highly relational data that must be strictly taken care of to ensure academic accuracy and data integrity.

3.4.1.1 Database

The GradMap system shall utilize a relational database architecture to manage the complex, interconnected nature of academic scheduling.

3.5 Design Constraints

3.5.1 Standards Compliance (O: Dani)

The GradMap system shall follow standard web development and security practices. It will use HTML, CSS, JavaScript, and MySQL following accepted industry standards. The system shall securely hash passwords using an industry-standard algorithm. The application will be compatible across major web browsers.

3.6 Software System Attributes

3.6.1 **Reliability**(O: Dani M:Jackson)

3.6.1.1 The system shall present information using a structured layout with labeled sections and navigation menus, allowing all users to access key features.

3.6.1.2 The system shall use consistent fonts, colors, and formatting across all pages to improve readability.

3.6.1.3 The system shall display schedules, alerts, and feedback using standardized formats, where all text is readable at a minimum font size of 12pt and alerts are displayed.

3.6.1.4 The system shall ensure all text and interface elements are readable across different devices and screen sizes.

3.6.2 Availability (O: Ivan)

3.6.2.1 The system shall be available to all users during standard usage periods

3.6.2.2 The system shall support concurrent access by multiple users

3.6.2.3 The system shall ensure that core features remain accessible:

- Dashboard viewing
- GradPlan management
- Course browsing
- Advisor review

3.6.2.4 The system shall allow recovery from system failures via restart

3.6.2.5 The system shall minimize downtime during operation

3.6.3 Security (O: Brice)

3.6.3.1 The system shall securely authenticate all users, utilizing JSON Web Tokens (JWT) to establish and manage user session upon successful login.

3.6.3.2 The system shall never store passwords in plaintext. All user passwords must be securely hashed and salted using the bycrypt algorithm prior to insertion into the Users database table.

3.6.3.3 The system shall enforce strict Role-Based Access Control (RBAC). The application must verify the user's role (Student, Advisor, or Administrator) embedded within their JWT to ensure they can only access pages, API endpoints, and data explicitly permitted for their profile type.

3.6.4 Maintainability (O: Matthew)

3.6.4.1(M: Dani) The system shall operate as a web-based application with at least 99% uptime and shall require no more than two hours of maintenance per month for student and advisor users. Administrators shall be able to perform system updates related to course data within a maximum of 2 hours without impacting core system functionality.

3.6.4.2 The system shall remain accessible as long as stable internet connection is maintained.

3.6.4.3 The system shall require users to perform basic low maintenance updates on their device to maintain access to GradMap.

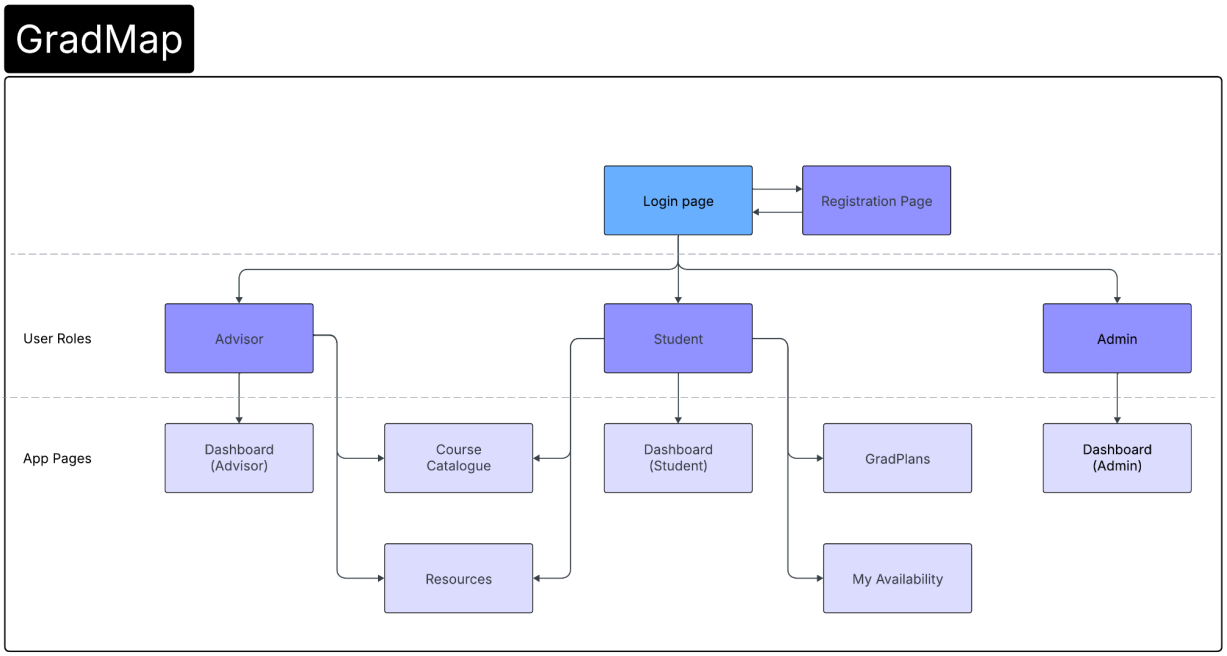
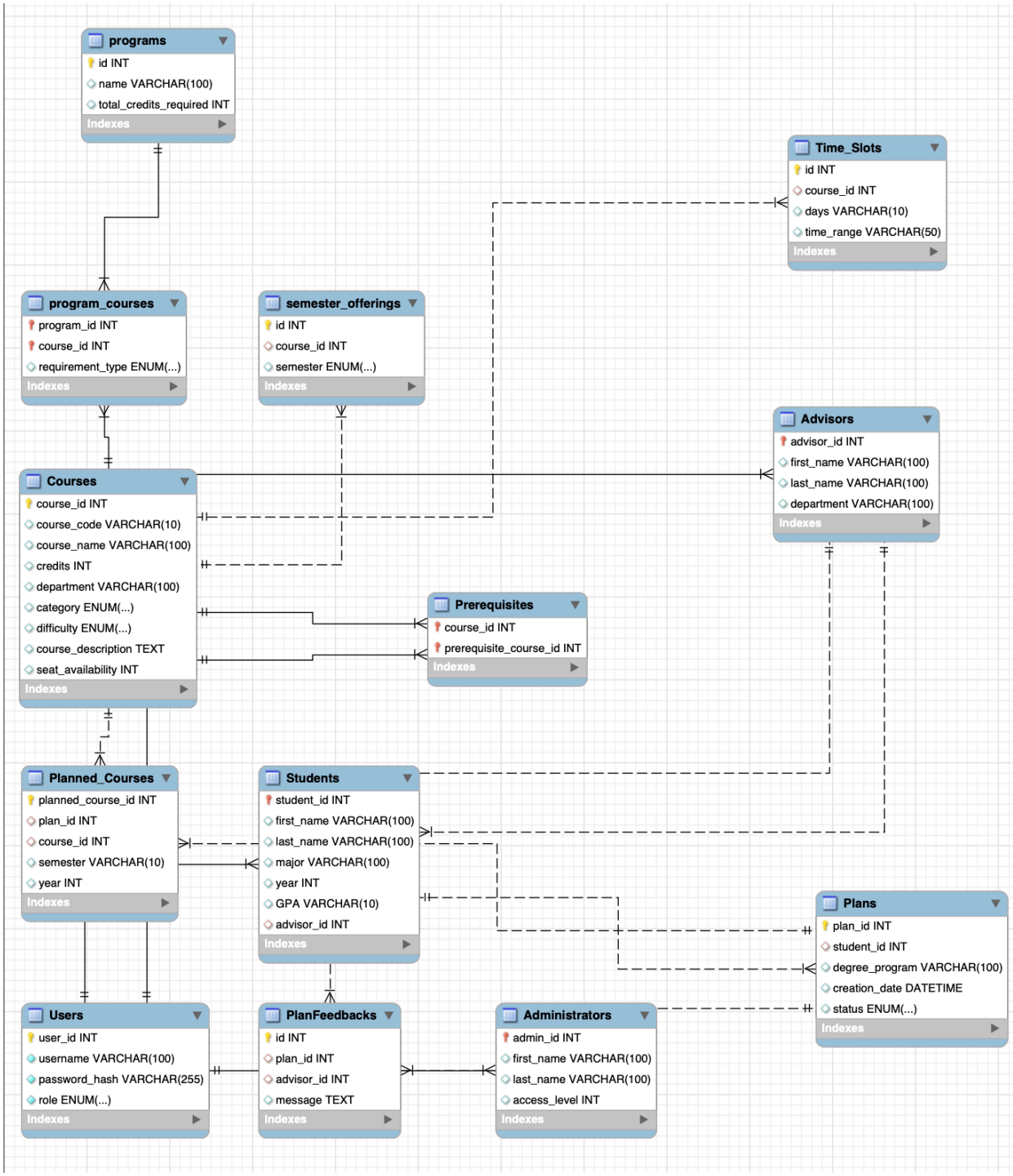
3.6.5 Portability (O: Jackson) (M: Matthew)

3.6.5.1 The system shall ensure that any and all Web Browsers can run the application.

3.6.5.2 The system shall be tested vigorously across any and all browsers to ensure that it runs smoothly for all users.

4 Appendices

4.1 Appendix A: Site Map



4.2 Appendix B: Database Schema